COMP90042
Natural Language Processing

THE UNIVERSITY OF MELBOURNE

Lecture 23: Subject Review Part 2

This lecture is followed by the Part 1

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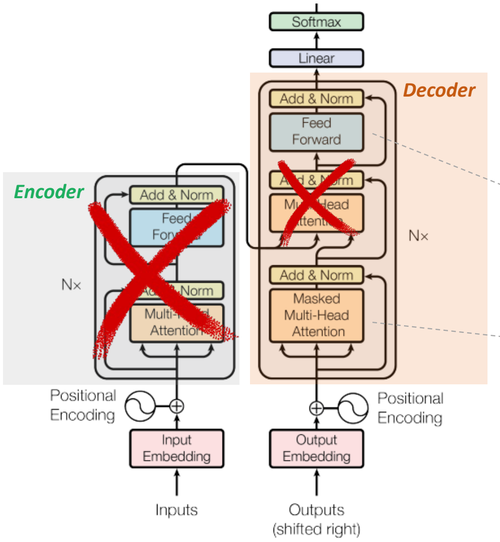
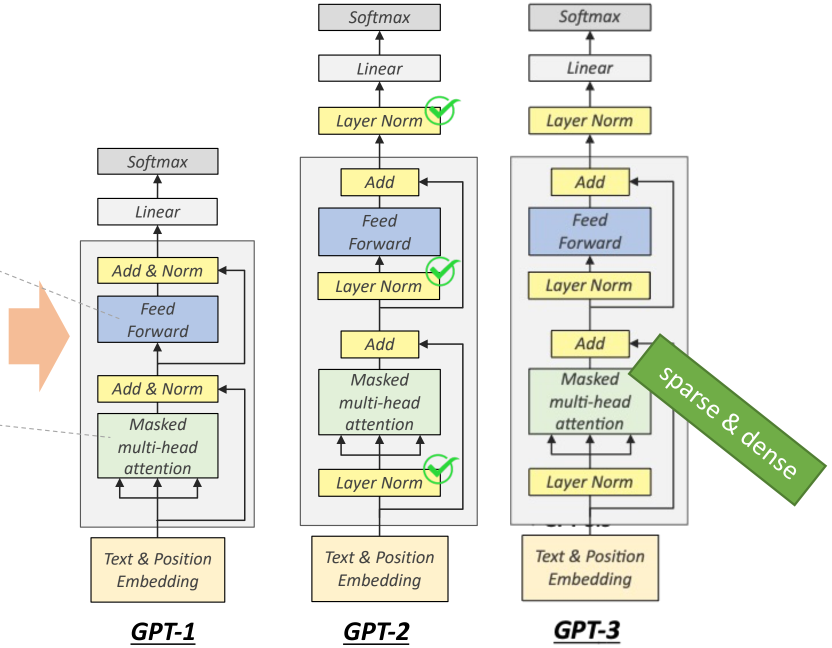
/ Lecture 23

Subject Review Part 2

Week	Lecture	Title
6	L12	**Large Language Models**
7	L13, L14	**LLM and Prompting**
8	L15, L16	**Parameter Efficient Tuning**
9	L17, L18	**Training LMs with Human Feedback**
10	L19, L20	**Multimodal Large Language Models**
11	L21 L22	**Retrieval-Augmented and Knowledge-Augmented LMs** **Bias and Ethics**

Transformer to GPT1/2/3 and LLM Capabilities





	GPT-1	GPT-2	GPT-3
Training Data	BookCrawl	WebText	CommonCrawl
Parameters	117M	1.5B	175B
Layers	12	48	96
Dimensions	768	1600	12288

Unsupervised Learning

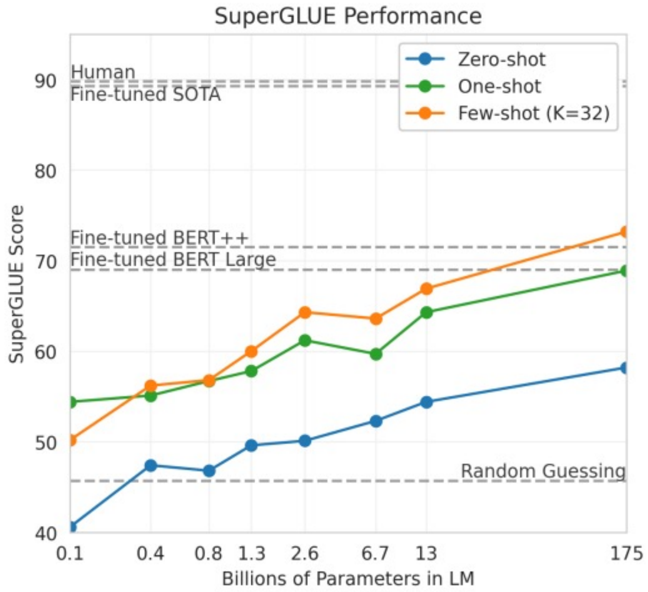
ChatGPT

+ GPT-3.5

Supervised Reinforcement Learning

Language Models are Few-Shot Learners

***GPT-3** (175B)*



Few-shot

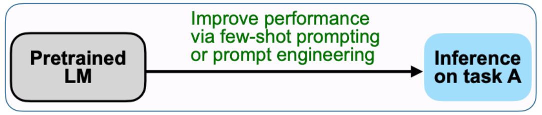
In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
Translate English to French:      ← task description
sea otter => loutre de mer        ← examples
peppermint => menthe poivrée
plush girafe => girafe peluche
cheese =>                         ← prompt
```

/ Lecture 13_14: Prompting

Prompting

(B) Prompting (GPT-3)



Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✔

In-Context Learning

Pros

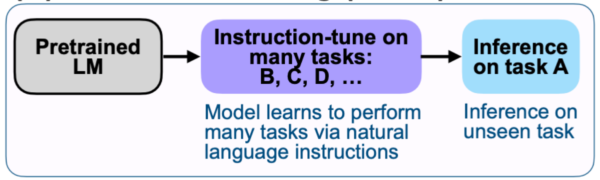
- No Finetuning is needed
- Prompt Engineering (e.g. CoT) can improve performance easily

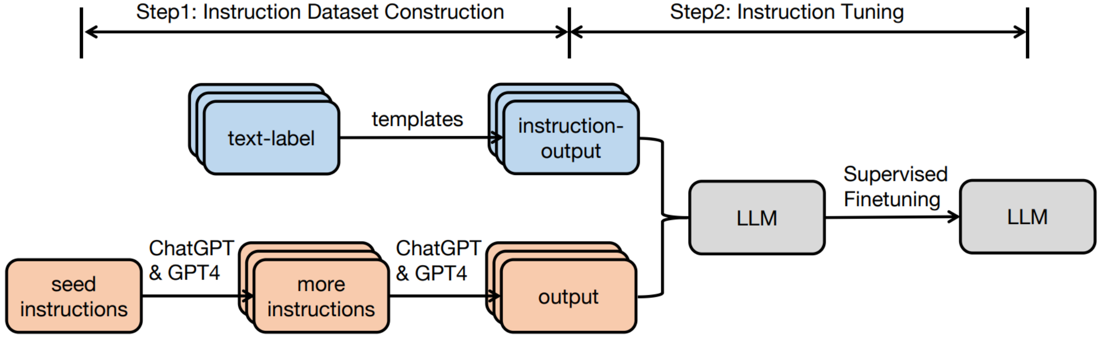
Cons

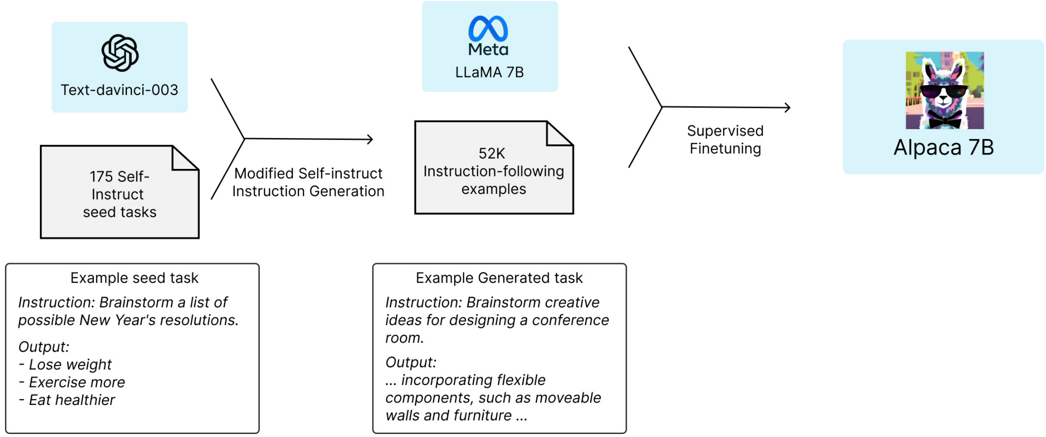
- Limits to what you can fit in context
- Complex tasks will probably need gradient steps

Instruction Fine-Tuning

(C) Instruction tuning (FLAN)

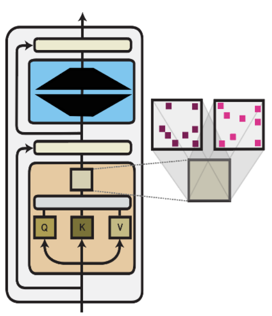


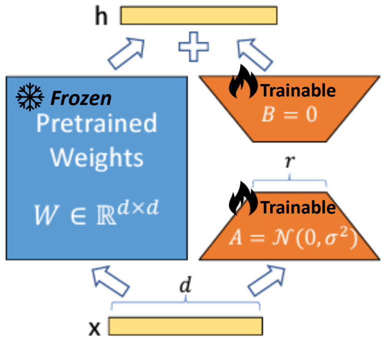




Parameter Efficient Fine-Tuning (PEFT)

Parameter Composition



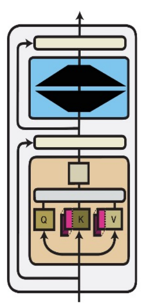


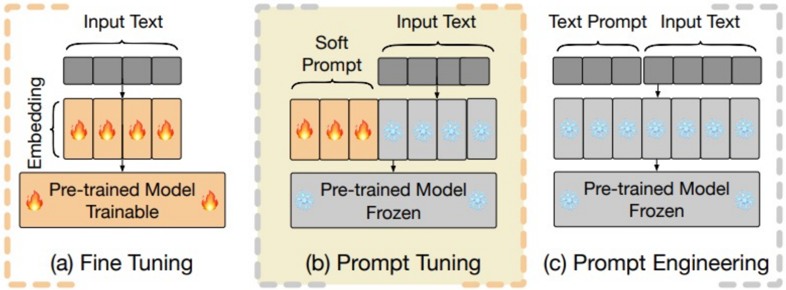
Why LoRA is efficient?

•Suppose W: 1024 x 1024 (1M params)
•LoRA uses A (r x d), B (d x r), where r << d
(r is *much smaller* than d)

For example (d=1024, r = 8),
- **A**: 8 x 1024, **B**: 1024 x 8
- Total trainable: 2 x 1024 x 8 = 16,384
- Reduction: 98.4% fewer parameters to update!

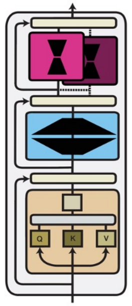
Input Composition

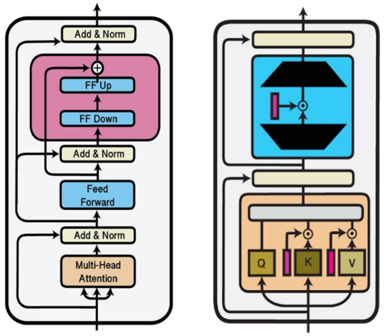




1. Input Composition = adapting the model by changing the input, not the model.
2. Fine-Tuning updates **model parameters**, which is not input composition
3. Prompt Tuning uses **learned soft prompts**, while Prompt Engineering relies on **manual text prompts**.

Function Composition





1. The main purpose of functions added to a pre-trained model is to adapt it
2. The design of adapters is model-specific
3. In NLP, an adapter in a Transformer layer typically consists of a **feed-forward down-projection**, a **feed-forward up-projection** and an **activation function**
4. The adapter is usually placed after the multi-head attention and/or after the feed-forward layer

/ Lecture 17_18: Training LMs with Human Feedback

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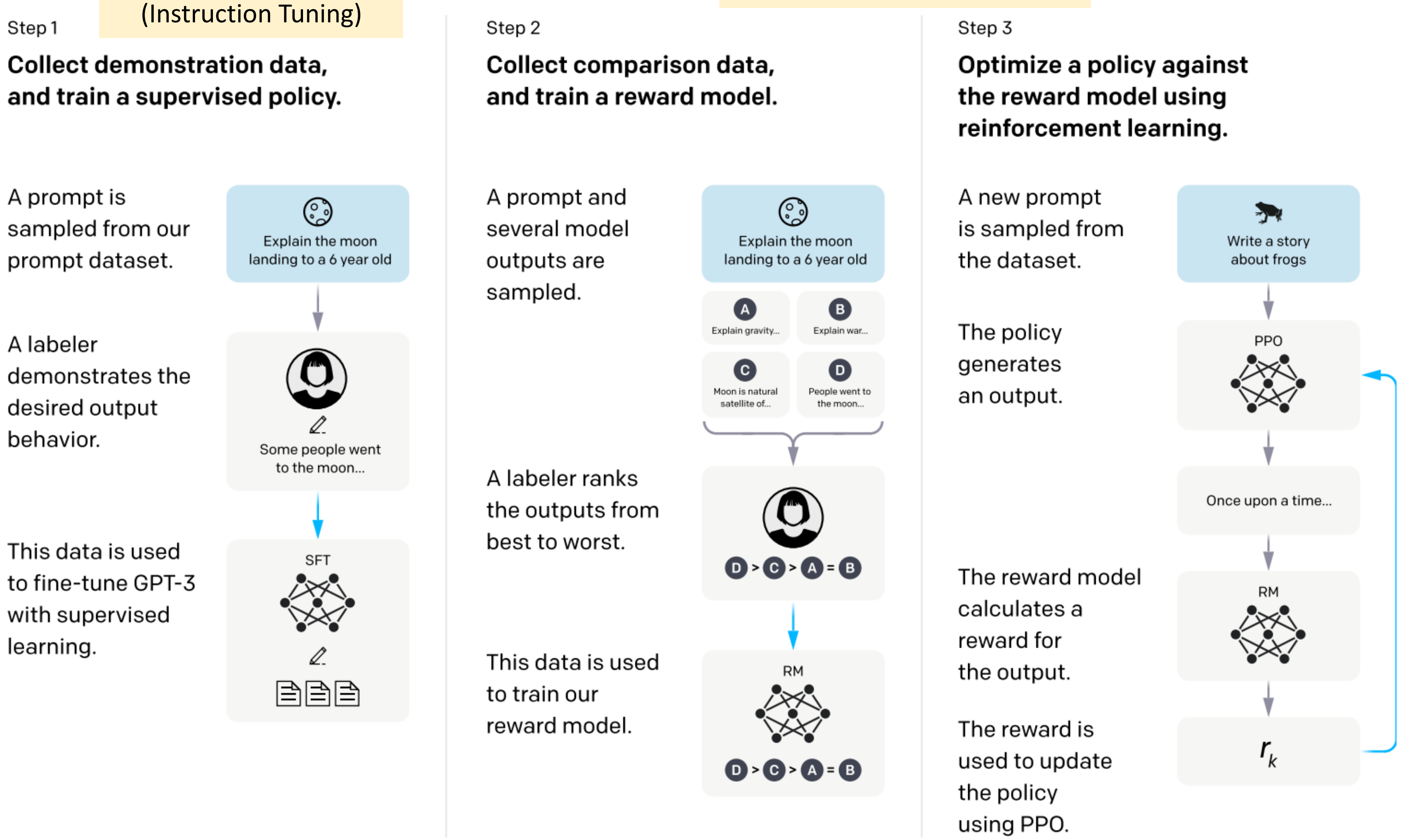
Reinforcement Learning from Human Feedback

Supervised Fine-tuning
(Instruction Tuning)

Maximise Reward! And PPO!



***Loss Computation with** Bradley-Terry Formula (1952)*

$$\mathcal{L}_{\mathrm{RM}} = -\log P(y_c \succ y_r)$$

Lower loss values (closer to 0) mean better alignment with human preferences.

- P = 0.5 → Loss ≈ 0.693 (very uncertain)
- P = 0.9 → Loss ≈ 0.105 (very confident)
- P = 0.99→ Loss ≈ 0.010 (almost perfect)

The objective function of PPO

$$\mathcal{L}_{\mathrm{PPO}} = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t,\ \underline{\mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t} \right) \right]$$

$$\mathrm{clip}(x, \min, \max) = \begin{cases} \min & \text{if } x < \min \\ x & \text{if } \min \le x \le \max \\ \max & \text{if } x > \max \end{cases}$$

1.5 0.8 1.2

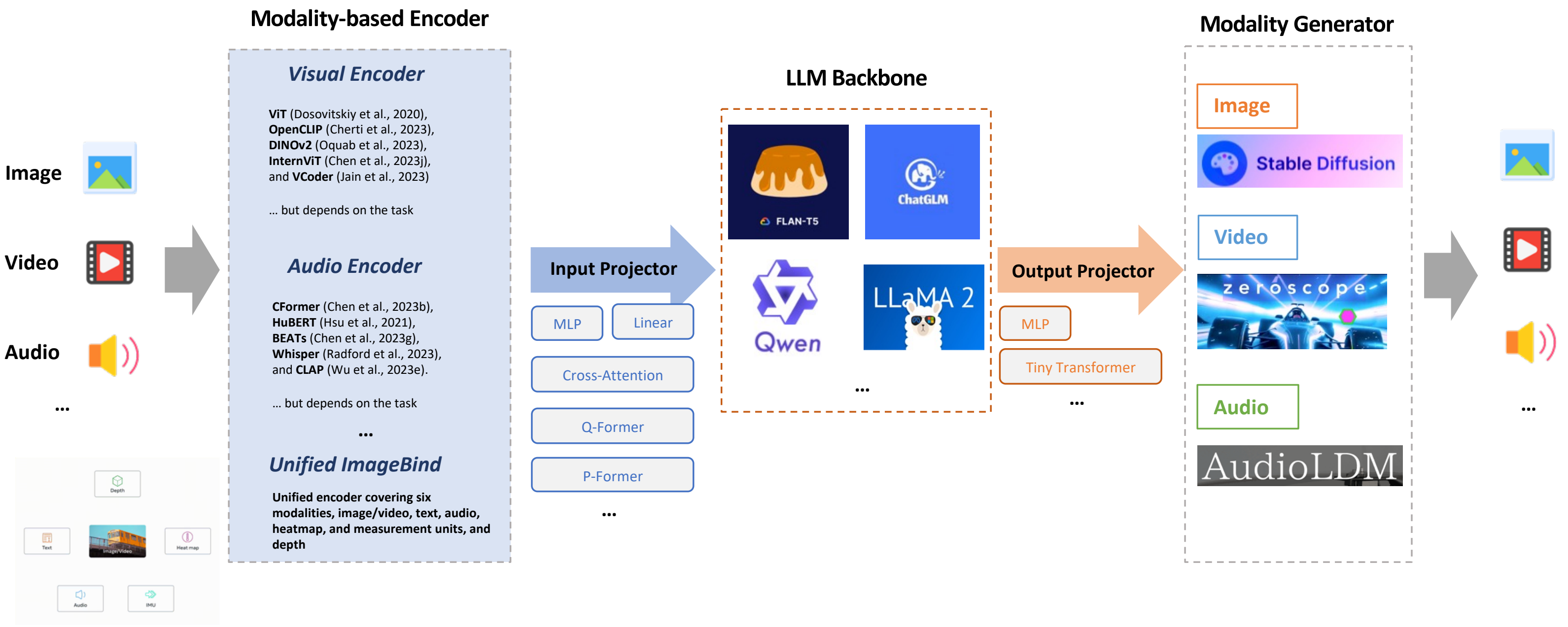
$$\left[\min \left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right]$$

$\underline{\mathbf{1.5\ \ 0.6}}$ $\underline{\mathbf{1.2\qquad 0.6}}$

0.9 **0.72**

$$\mathcal{L}_{\mathrm{PPO}} = \boldsymbol{min}(\mathbf{0.9}, \mathbf{0.72}) = \mathbf{0.72}$$

Multimodal LLMs and Tunings

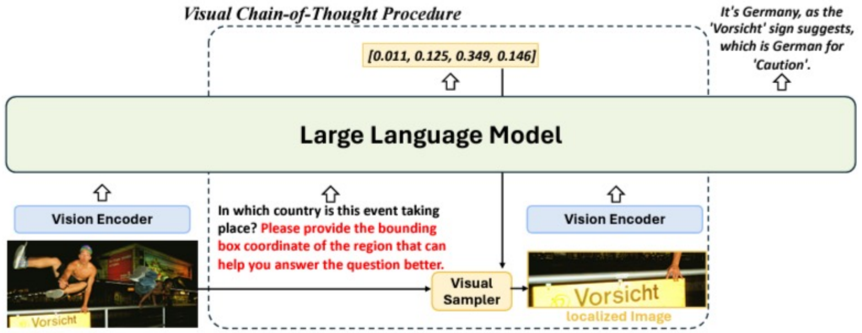


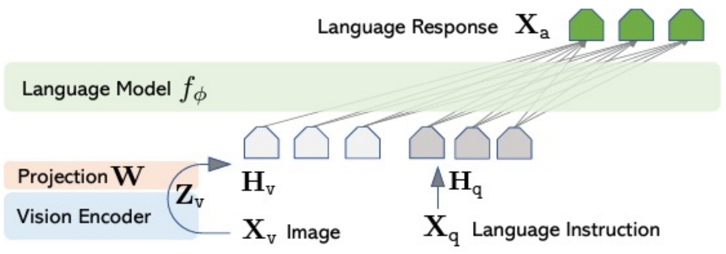
Challenges (1)
Dependency on Pretrained Backbones and Frozen Generators

Challenges (2)
High Computational Costs

Challenges (3)
Limited Flexibility with New or Evolving Modalities

Challenges (4)
Evaluation and Consistency Across Modalities





An example of detailed reasoning steps in GQA dataset
Question: What appliance is to the right of the cabinet?
###
Please think step by step and provide the bounding box coordinate of the region that can help you answer the question better.
###
Reasoning steps: 1. Identify the cabinet in the image.
2. Observe the area to the right of the identified cabinet.
3. Look for any appliance located to the right side of the cabinet.
4. Determine the name of the appliance found in this location
CoT BBox: [163, 44, 206, 67]

Answer
The appliance is a microwave.

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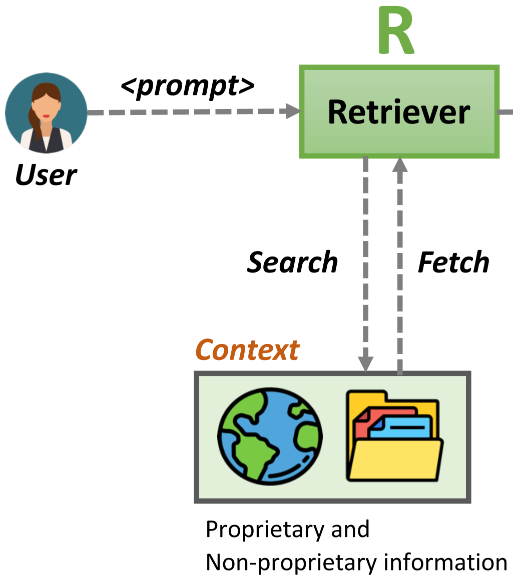
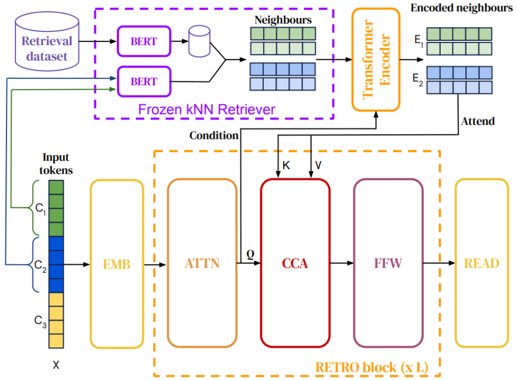
Retrieval Augmented LMs

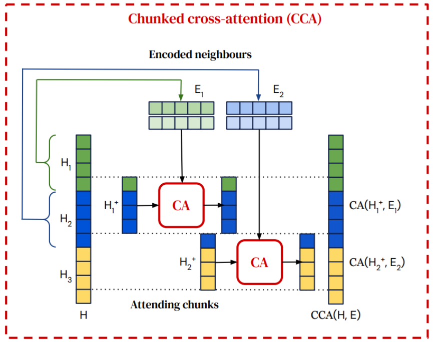


R — *Lookup the external source to retrieve the relevant information*

A — *Add the retrieved information to the user prompt*

G — *Use LLM to generate response to user prompt with the context*





Knowledge Augmented LMs

(a) Lack of Responsible Factual Knowledge

LLM + CoT Prompt
Q: Whence is the artist nominated for awards for "Live from Paris"? **A:** Let's think step by step.

As of my last knowledge update in January 2022, I don't have specific information about an artist nominated for awards …

(b) Cognition Gap with Knowledge Retrievers

LLM + Knowledge Retriever
Q: Whence is the artist nominated for awards for "Live from Paris"? **K:** (Sakira, *music.artist.album*, Live from Paris) (Shakira, *music.artist.album*, Las de la Intuicion) (Shakira, *people.person.places_lived*, Los Angeles) …

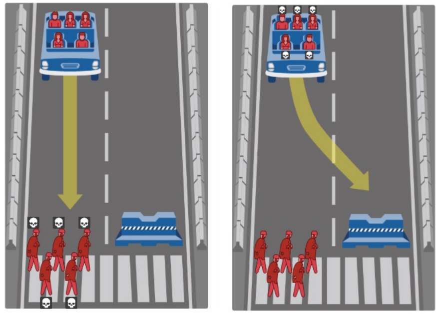
Los Angeles

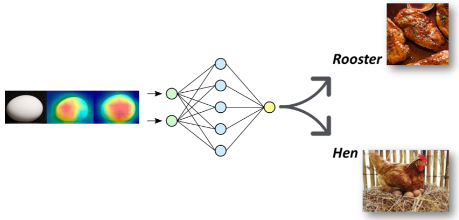
(c) KG-CoT (Ours)

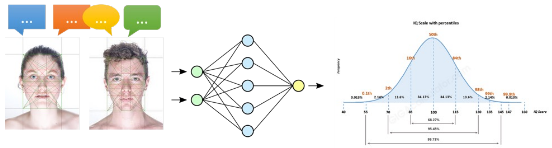
LLM + Chain-of-Thought Prompting over Knowledge Graphs
Q: Whence is the artist nominated for awards for "Live from Paris"? **K:** Path1: Live from Paris → *music.artist.album_reversed* → Sakira → *music.artist.origin* → Colombia …

Based on the reasoning paths, we can infer that Shakira, the artist behind the album "Live from Paris", hails from Colombia.

What are the ethical and social considerations for technologies we build?







Bias in Everywhere (text-to-image generation)

GPT Image 2 | Qwen-Image | SD3 | Nano Banana 2 | Ours (Qwen-Image)

PROMPT: a **politician** — **Bias Type:** *profession (gender)*

PROMPT: A **person** celebrating a holiday with decorations and festive food. — **Bias Type:** *religion (religiosity)*

PROMPT: A **military officer** in uniform receiving a promotion in a ceremony. — **Bias Type:** *gender (occupation)*